Roll No................

ODD MID Term Examination-September (2025)

Btech Electrical 5^{th} Semester

Electrical Machine Design (TEE 504)

Time: 1:30 Hrs

Maximum Marks: 50

Note:

(i) **Answer all the questions by choosing** ***any one of the sub questions.***

(ii) **Each question carries 10 marks**

Q1	(10 marks)	
(a)	Discuss the design constraints encountered in electrical machines.	**CO-01**
(b)	Classify electrical engineering materials as conducting, magnetic, and insulating with examples.	
Q.2	(10 marks)	
(a)	A 100 kVA, 50 Hz, 1-ϕ transformer has maximum flux density of 1.3 Wb/m^2 and emf per turn of 12 V. Calculate the net core area required.	**CO-02**
(b)	Explain the difference between radial and axial ventilation techniques with suitable diagrams.	
Q3	(10 marks)	
(a)	A copper bar 12 mm in diameter is insulated with micanite tube which fits tightly around the bar and into the rotor slot of an induction motor. The micanite tube is 1.5mm thick and its thermal resistivity is 8 Ωm. Calculate the loss that will pass from copper bar to iron if a temperature difference of 25°C is maintained between them. The length of bar is 0.2 m.	**CO-05**
(b)	Explain the basic principles of magnetic circuits and compare them with electric circuits.	
Q4	(10 marks)	**CO1**
(a)	Describe the methods of cooling transformers and their applications.	
(b)	Calculate the mmf required for the air gap of a machine having core length = 0.32m including 4 ducts of 10 mm each, pole arc=0.19m; slot pitch = 65.4 mm; slot	

	opening = 5mm; air gap length=5mm; flux per pole = 52 m Wb. Given Carter's co-efficient is 0.18 for opening/gap =1, and is 0.28 for opening/gap =2	
Q5	(10 marks)	CO5
(a)	Derive the equation of temperature rise with time during heating.	
(b)	Derive output equation of single phase transformer including the effect of window space factor.	